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بسم الله الرحمن الرحيم

The derivatives
المشتقات

متغير تابع — $y = f(x)$ — متغير مستقل

1) explicit form
دالة صريحة

ex $y = f(x) = 3x+1$
$f(0) = 1$

2) Implicit fn. الدالة الضمنية

$f(x,y) = 0$

ex $x\sin(x+y) + 7 = 0$

3) parametric fns
الدوال البارامترية

ex $x = \sin\theta$, $y = \cos\theta$

ex. $x = t^2+1$ $y = \sqrt{t}$

Derivatives
مشتقات

$y = f(x)$

1st derivative المشتقة الأولى

$$\lim \frac{f(x+h) - f(x)}{h}$$

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Derivatives
مشتقات

$y = f(x)$

1st derivative المشتقة الأولى

$$y' = f' = \frac{dy}{dx} = \frac{df}{dx} = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

ex find the 1st derivative of x^2 by using the Def.
find y' for $y = x^2$

Sol

$f(x) = x^2$

$$y' = f' = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$f(x) = x^2$

$f(x+h) = (x+h)^2$

$$= \lim_{h \to 0} \frac{(x+h)^2 - x^2}{h}$$

$$= \lim_{h \to 0} \frac{\cancel{x^2} + 2x\cancel{h} + h^2 - \cancel{x^2}}{\cancel{h}} = \lim_{h \to 0} (2x + h)$$

$$= \boxed{2x}$$

* not all the continuous fns are differentiable
ليس كل الدوال المتصلة قابلة للاشتقاق

but all differentiable fns are continuous
لكن كل الدوال القابلة للاشتقاق متصلة

ex prove that $f(x) = |x|$ is continuous but not differentiable

* Rules of derivatives

① $\frac{d}{dx}(\underset{\text{ثابت}}{c}) = 0$

② $\frac{d}{dx}(f(x) \pm g(x)) = \frac{df}{dx} \pm \frac{dg}{dx} = f' \pm g'$

③ $\frac{d}{dx}(c_1 f(x) + c_2 g(x)) = c_1 f' + c_2 g'$

④ $\frac{d}{dx} x^n = n x^{n-1}$

⑤ $\frac{d}{dx}(f \cdot g) = f g' + f' g$
$= f' g + g' f$

⑥ $\frac{d}{dx}\left(\frac{f}{g}\right) = \frac{f' g - g' f}{g^2}$

⑦ $\frac{d}{dx}\sqrt{x} = \frac{1}{2\sqrt{x}}$. $\frac{d}{dx} x^{\frac{1}{2}} = \frac{1}{2} x^{\frac{1}{2}-1} = \frac{1}{2} x^{-\frac{1}{2}} = \frac{1}{2\sqrt{x}}$

⑧ $\frac{d}{dx}(f(x))^n = n(f(x))^{n-1} f'(x)$

⑨ $\frac{d}{dx}\sqrt{f(x)} = \frac{f'(x)}{2\sqrt{f(x)}}$

$= \frac{d}{dx}(f(x))^{\frac{1}{2}} = \frac{1}{2}(f(x))^{\frac{1}{2}-1} \cdot f'(x)$

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$= \frac{1}{2}(f(x))^{-\frac{1}{2}} \cdot f'$

$= \frac{1}{2\sqrt{f(x)}} \cdot f'$

$$= \frac{1}{2}(f(x))^{-\frac{1}{2}} \cdot f'$$

$$= \frac{1}{2\sqrt{f(x)}} \cdot f'$$

find y' for.

① $y = 3x^2 + 7x + \sqrt{x} + 3$

$y' = 6x + 7 + \frac{1}{2\sqrt{x}} + 0$

② $y = (2x^2+1)^{10}$

$y' = 10(2x^2+1)^9 \cdot 4x$

③ $y = \sqrt{x^3+5}$

$y' = \frac{1}{2\sqrt{x^3+5}}(3x^2)$

$\frac{d}{dx}(fg) = f'g + fg'$

④ $y = (2x^2+1)(5x^{10}+2x)$

$y' = 4x(5x^{10}+2x) + (2x^2+1)(50x^9+2)$

⑤ $y = \frac{2x+1}{x^3+5}$

$y' = \frac{2(x^3+5) - 3x^2(2x+1)}{}$

الاشتقاق

find y' for:

$$y^2 + 5y + 7x = 0$$

بالاشتقاق بالنسبة لـ x

$$2yy' + 5y' + 7 = 0$$

$$= (2y+5)y' = -7$$

$$\therefore y' = \frac{-7}{2y+5}$$

$(xy^2)'$

مشتقة حاصل ضرب دالتين

$$= 1(y^2) + x(2yy')$$

$$= y^2 + 2xyy'$$

② $y^2 + xy^2 + \sqrt{x} + \sqrt{y} = 0$

بالاشتقاق بالنسبة لـ x

$$2yy' + y^2 + 2xyy' + \frac{1}{2\sqrt{x}} + \frac{y'}{2\sqrt{y}} = 0$$

$$\Rightarrow \left(2y + 2xy + \frac{1}{2\sqrt{y}}\right)y' = -\left(y^2 + \frac{1}{2\sqrt{x}}\right)$$

$$\Rightarrow y' = -\frac{\left(y^2 + \frac{1}{2\sqrt{x}}\right)}{2y + 2xy + \frac{1}{2\sqrt{y}}}$$

If $z = y^3 + 2y + 7 \longrightarrow z = z(y)$

find $z' = \frac{dz}{dy}$

$z' = \frac{dz}{dy}$